

WE MAKE MOTION *without* CODE *

How a 30-second motion piece and its percussive score were designed, choreographed, synthesised and rendered, from a one-paragraph brief to 4K masters of two editions.

30 S • 1920×1080 • 60 FPS • 120 BPM
2D AND 3D EDITIONS • 1,800 FRAMES EACH
*THIS DOCUMENT IS ALSO CODE.

At a glance

The brief asked for a sleek, fluid, 30-second motion sequence featuring five words (**We, Make, Motion, Without, Code**) with percussive sound and distinct effects tied to key moments. The result is a short film that argues its headline for 26 seconds and then footnotes itself.

Everything was built in code: a single HTML page draws each frame on a canvas as a pure function of time, and the soundtrack is synthesised from oscillators and filtered noise by the same page. No video editor, no audio workstation, no samples, no stock assets. One cue sheet drives both picture and sound, which is how every bounce, slam and wipe lands on its hit.

DELIVERABLES

MASTERS `we-make-motion-4k.mp4` (3840×2160, 39.5 MB) and `we-make-motion.mp4` (1920×1080, 15.9 MB), 60 fps, H.264, AAC 256 kb/s.

INTERACTIVE PLAYER A private claude.ai artifact: real-time playback, waveform scrubber, dope sheet, and a Principles overlay that names the technique on screen.

SOURCE Repository `digbyhogan/main`, folder `motion/we-make-motion/`, with an offline renderer that reproduces the master.

3D EDITION `we-make-motion-3d-4k.mp4` (91.4 MB at 24 Mb/s), `we-make-motion-3d.mp4` (1080p, 29.1 MB) and a live WebGL player: the same cut and score with extruded type, glow, grit and mattes.

THIS DOCUMENT Approach, workings, and the progression from first pass to final.

BY THE NUMBERS

Duration	30.000 s · 15 bars of 4/4
Tempo	120 BPM · beat = 0.5 s
Frames	1,800 at 60 fps
Sub-frames rendered for motion blur	43,200 (24 per frame)
Scenes	8
Particles forming “WE”	1,897
Sound sources scheduled	686 oscillators + noise
Samples or recordings used	0
Loudness · peak	−16.4 LUFS · −0.8 dBFS
Colours · typefaces · protagonists	5 · 3 · 1 dot
Lines in the film’s page	1,941
4K render time (4 CPU cores)	72 min

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The brief, and the prompt I wrote myself

WHAT WAS ASKED

“Self prompt and create a modernistic, sleek and fluid motion design sequence that really shows off your capabilities, not just in design but in the principles of motion and motion graphics ...feature some key words: We, Make, Motion, Without, Code ... 30 seconds, a percussive sound and distinctive sound effects that align with key moments.”

HOW I READ IT

Three things were being tested at once: visual design, the craft of animation (the twelve classic principles and the modern motion-graphics vocabulary), and audio-visual sync. A pretty title sequence would only cover the first. So the piece had to *demonstrate* principles on screen, legibly, and every one of them had to be timed to sound.

The five words also carry an irony: a film saying “without code” that is built entirely from code. I decided to use that rather than hide it. The film makes the claim with full conviction, then adds an asterisk.

ASSUMPTIONS I MADE INSTEAD OF ASKING

FORMAT 16:9 at 1920×1080 and 60 fps, because fluid motion reads better at 60.

OUTPUT An MP4 master plus a live, scrubbable web version.

SOUND Percussion-led, with tuned accents in C major pentatonic so plucks and pings sit together.

THE SELF-PROMPT

```
// creative brief, written before any code
Make a 30-second film that says
"We make motion without code",
built entirely from code, and let it notice.
```

Rules

1. One protagonist: a flamingo dot. It opens the film, appears in almost every scene, and closes it.
2. One tempo: 120 BPM. Every cut, hit and transition lands on the 16th-note grid.
3. One cue sheet: picture and sound read the same timestamps, so sync can't drift.
4. Every frame is a pure function of time, so it can be scrubbed, rendered offline and motion-blurred without re-simulation.
5. Each word gets its own motion language: WE assembles, MAKE constructs, MOTION flows, without is still, CODE decodes and collapses.
6. Show the principles, don't list them.
7. Restraint: five colours, three typefaces, flamingo belongs to the dot.
8. End on the joke. Keep it dry.

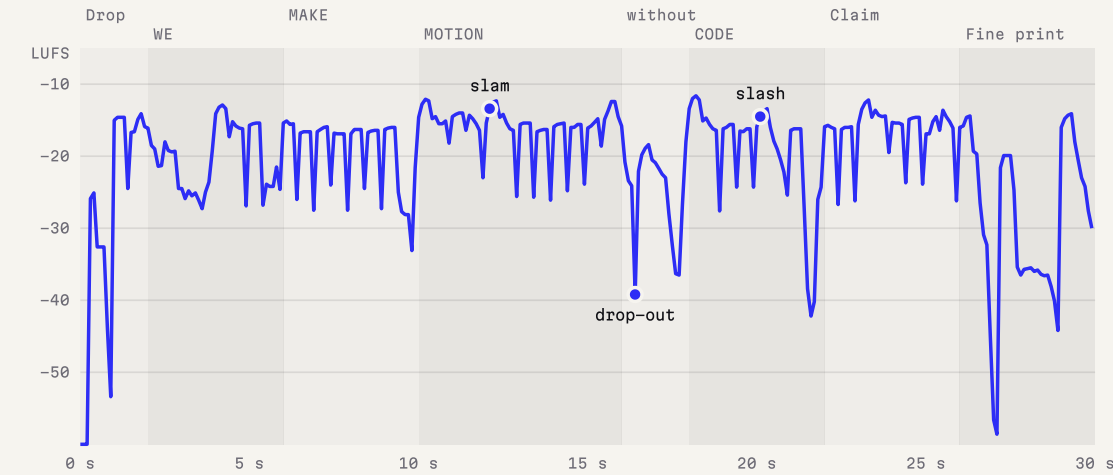
The rules above shaped every later decision. Rule 4 is the technical one that everything else depends on. Two rules bend in practice: a few timings sit off the grid (page 6), and flamingo is lent to a few accents, such as MOTION's beat flashes and the code numerals.

A claim in eight beats, and one footnote

TIME	BEAT	MOTION LANGUAGE
0:00	The drop	A dot falls and bounces. The film introduces its protagonist and its tempo.
0:02	WE	One becomes many: mitosis, then a particle swarm that assembles the word.
0:06	MAKE	Construction: a grid, measurements, letters drawn stroke by stroke.
0:10	MOTION	Flow, then excess: a wave, a marquee, a build that collapses to a line.
0:16	without	Absence: silence, stillness, a thin serif. The camera dives into the ink.
0:18	CODE	Decoded, struck through, dropped.
0:22	The claim	All five words, each entering with its own verb.
0:26	The fine print	The full stop becomes an asterisk. The footnote types itself.

The dot is in almost every scene: it drops and bounces, splits into the particles, lands in the counter of the A, surfs the wave, becomes the tittle of the i, returns as the full stop, turns into the asterisk, and finally shrinks back to its opening size and blinks out at the centre of the frame.

ENERGY ACROSS THE FILM



Momentary loudness of the final soundtrack (EBU R128, 400 ms window) sampled every 100 ms. Shaded bands are the eight scenes. The shape is deliberate: two builds, a sharp dip at 16 s, a second act, a hush, and one last hit at 29 s before a short fade.

Five colours, three typefaces, one dot

PALETTE



Ink #111116	Dark ground (drop, WE, CODE, fine print) and type on light scenes. Slightly blue-black, never pure black.
Chalk #F1EFE9	Light ground (MAKE, without) and type on dark and blue scenes. Warm off-white.
Ultramarine #2D2DF5	The loud ground: MOTION and the claim. Electric, close to International Klein Blue.
Flamingo #FF5A87	The dot and its deeds: echo trails, the strike-through, the asterisk. Lent out for a few accents: MOTION's beat flashes, code numerals, glitch bars.
Butter #FFD34E	Code only: brackets, selection highlights, the caret.

I avoided the common defaults (near-black with a single acid accent, cream with terracotta). Three grounds alternate scene to scene, giving the cut rhythm a colour rhythm too: ink → ink → chalk → ultramarine → chalk → ink → ultramarine → ink.

TYPOGRAPHY

MOTION

Unbounded 900. Wide, geometric and heavy. The shouting voice: WE, MOTION, the claim.

without

Bodoni Moda Italic. A high-contrast Didone whisper. Used for one word only, so the contrast lands like a change of tone of voice.

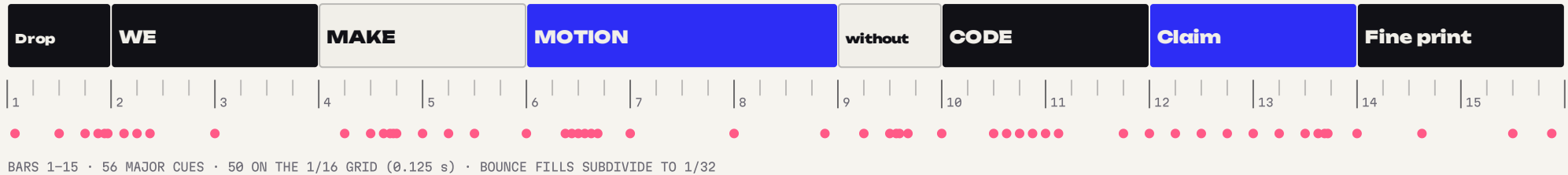
<CODE />

Martian Mono. A wide monospace that rhymes with Unbounded's proportions. CODE, the source code backdrop, the HUD and the footnote.

MAKE uses none of them: its letters are a custom monoline alphabet built from 13 stroke segments with round caps and joins, so they can be drawn on like a pen.

A thin **HUD slate** frames most scenes: chapter label, tempo, running SMPTE timecode, bar number and a four-square beat indicator, so the viewer can literally see the beat.

Everything lands on a grid



The film as 15 bars of 4/4 at 120 BPM. Each tick is a beat (0.5 s); flamingo marks are the major hits read from the cue sheet.

ONE CUE SHEET, TWO CONSUMERS

Every hit lives once, in a single object called **CUE**, and the wipes, glitch windows and camera impacts are derived from it. The picture reads it to decide when a word locks or a letter decodes, and the score reads the same values to decide when a kick fires. If I move the slam from 12.0 s to 12.25 s, the camera punch, the flash, the scale spring, the kick, the snare, the clap and the crash all move together. Sync cannot drift because there is nothing to drift apart.

The same idea drives the camera: a list of **[time, strength]** impacts feeds a decaying shake and a zoom punch, so the frame physically reacts to the loudest hits.

WHY A GRID

120 BPM gives a beat of exactly 0.5 s, an eighth of 0.25 s and a sixteenth of 0.125 s. At 60 fps those are 30, 15 and 7.5 frames. 50 of the 56 major cues sit exactly on it. The exceptions are deliberate: the opening pop (0.15 s) and the dive into the o (17.35 s) are timed by feel, and the bounce fills halve their spacing down to 1/32 notes (0.0625 s), as real drum fills do. That is why the piece feels musical even when nothing is on screen but a dot.

```
// src/core.js · the single source of truth (abridged)
const CUE = {
  dotPop: 0.15, launch: 0.45,
  bounces: [1.0, 1.5, 1.75, 1.875, 1.9375],
  split1: 2.25, split2: 2.5, burst: 2.75, weLock: 4.0,
  make: [6.5, 7.0, 7.5, 8.0], hop: 8.5, hopland: 9.0,
  motion: 10.0, slam: 12.0, build: 14.0, collapse: 15.75,
  without: 16.0, ping: 16.5, portal: 17.35,
  code: 18.0, locks: [19.0, 19.25, 19.5, 19.75],
  slash: 20.0, fall: 20.25, straighten: 21.5,
  final: [22.0, 22.5, 23.0, 23.5, 24.0],
  finalBounces: [24.5, 25.0, 25.25, 25.375, 25.4375],
  reflow: 26.0, asterisk: 27.25, typeStart: 27.6,
  fadeOut: 29.0, blink: 29.75,
  // ... plus weSquash, wipe1, grid, dotInA, lineOut, iris, letterLand,
  tittle, wipe2, curtain
};

// picture: const d = dropDot(t)      → reads CUE.bounces
// sound:   bounceKit(CUE.bounces)    → kicks on the same values
```

Physics you can scrub

RULE 4: TIME IN, FRAME OUT

Every position is a pure function of t , and almost all are closed-form. The one exception is the marquee's scroll: its speed is integrated once at load into a lookup table that each frame reads by t . That is what makes the timeline scrubbable in the browser, lets the offline renderer jump to any frame, and makes true motion blur possible: to blur a frame I sample 24 instants inside the shutter and average them.

Randomness is deterministic too. Particle angles, glitch slices and scramble glyphs come from an integer hash of an index, so frame 1,206 is identical on every render.

EASING, BY JOB

OUTEXPO / OUTQUART Entrances and draw-ons. Fast attack on the beat, long settle.

INQUART / INCUBIC Exits and the portal zoom. Accelerate away.

INOUTBACK The reflow. A small pull back before travelling (anticipation built into the curve).

OUTBACK Pops. A touch of overshoot so things arrive with confidence.

ANALYTIC SPRING

```
function spring(dt, zeta = 0.4, f = 2.2) {
  const w = 2 * PI * f;
  const wd = w * sqrt(1 - zeta * zeta);
  return 1 - exp(-zeta * w * dt) *
    (cos(wd * dt) + (zeta * w / wd) * sin(wd
    * dt));
}
```

An under-damped harmonic oscillator solved exactly. It gives overshoot and settle (follow-through) without any state. Used for the MOTION slam, the wave amplitude, the bracket slide-ins and the asterisk's arms, each with its own damping and frequency.

KEYFRAMED BALLISTICS

```
v0 = (y1 - y0) / T - g * T / 2
y(t) = y0 + v0 * (t - t0) + g * (t - t0)2 / 2
```

Instead of simulating gravity I specify where the dot must be and when, and solve for the launch velocity. Gravity stays physically constant (6,400 px/s²) while every landing hits its cue exactly.

THE BOUNCE IS A DRUM FILL

Bounce intervals halve: 0.5, 0.25, 0.125, 0.0625 s. Under constant gravity a bounce's height is $gT^2/8$, so heights quarter: 200, 50, 12.5, 3 px. The picture is physically right, and the sound is a textbook accelerating fill. The kick for each landing gets quieter and slightly higher-pitched, like a real ball on a real floor.

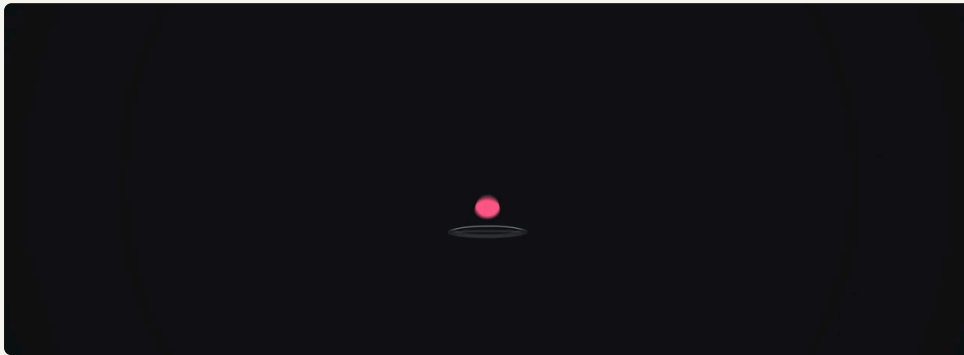
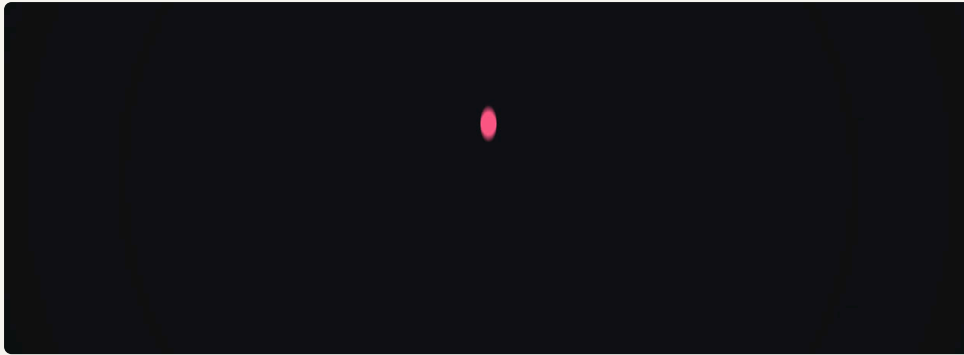
SQUASH & STRETCH, VOLUME KEPT

Stretch comes from speed ($1 + \min(0.45, |v|/5200)$) and squash from impact (a decaying cosine). Width is always the inverse of height, so the dot never gains or loses mass. The shape is anchored at its contact point so it squashes into the floor rather than floating above it. During the hop between A and E, the stretch is rotated to follow the velocity vector along the arc.

ECHOES, NOT JUST BLUR

MOTION's letters leave five flamingo echoes sampled 28 ms apart. They trace the letters' arcs and close up as the letters slow, so the trail length is itself a speedometer.

● 0:00-0:02 • THE DROP



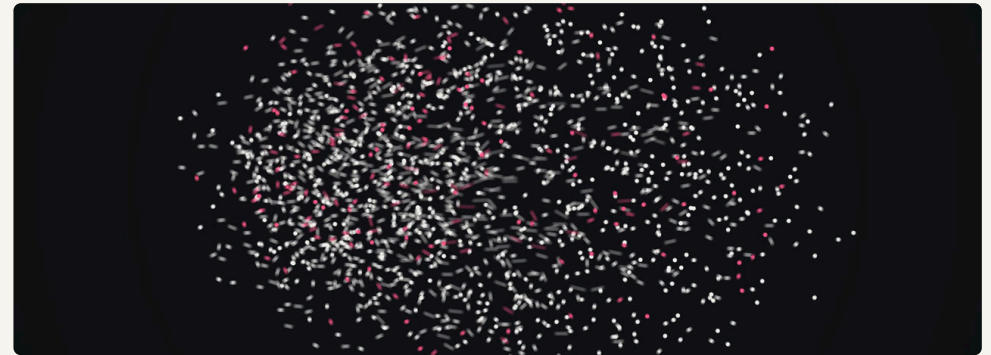
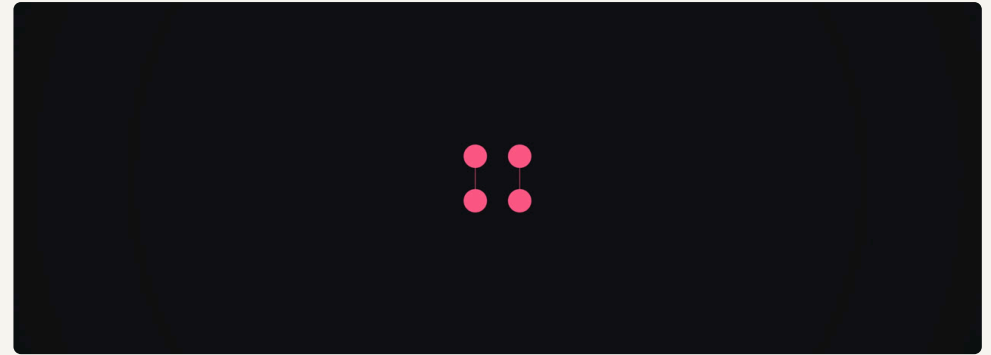
0.90 s falling, stretched along its velocity • 1.00 s impact:
squash, ripple, ground line

Picture. A dot pops in on a back ease (0.15 s), crouches (anticipation, 0.30–0.45 s), hops, and falls under gravity to land exactly on beat 3. Each landing throws an elliptical ripple across an implied floor, and the first draws a hairline horizon that fades out.

Sound. A G6 blip for the pop, a rising sine for the jump, then five kicks with clacks on the landings: 1.0, 1.5, 1.75, 1.875, 1.9375 s. The ball plays its own fill into the downbeat of bar 2.

TIMING ANTICIPATION SLOW IN/OUT SQUASH & STRETCH

● 0:02-0:06 • WE



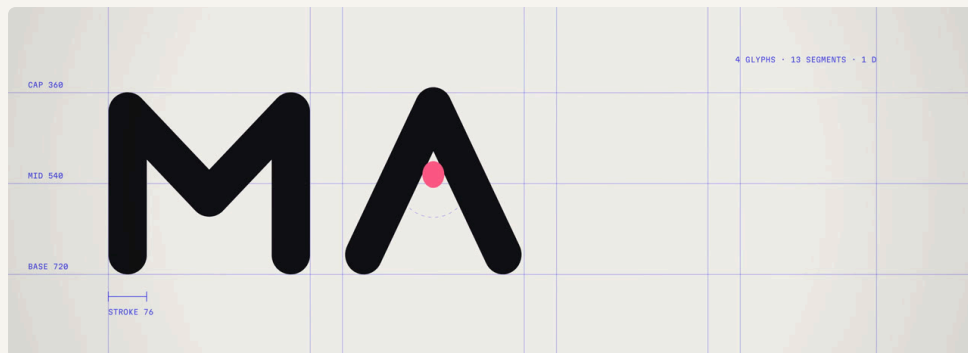
2.60 s mitosis, four cells • 3.40 s swarm curving home, left to right

Picture. The dot divides on successive eighth notes, $1 \rightarrow 2 \rightarrow 4$, necks pinching like cells. At 2.75 s it bursts into 1,897 particles. Their targets were sampled from the word WE rasterised offscreen. Each particle flies out on a swirling elliptical orbit, then homes in along an arc, launched in a left-to-right stagger so the word appears to write itself. At 4.0 s the dots flash into solid type with a flamingo outline echo, then the camera pushes in slowly. The letters crouch and leave upward while a liquid chalk wipe rises.

Sound. C5 and G5 plucks on the splits, a burst, 26 pentatonic blips scattered across the particles' flight, a riser, then kick, snare, clap and crash on the lock. The groove starts here.

SECONDARY ACTION ARCS STAGGER STAGING

● 0:06-0:10 • MAKE



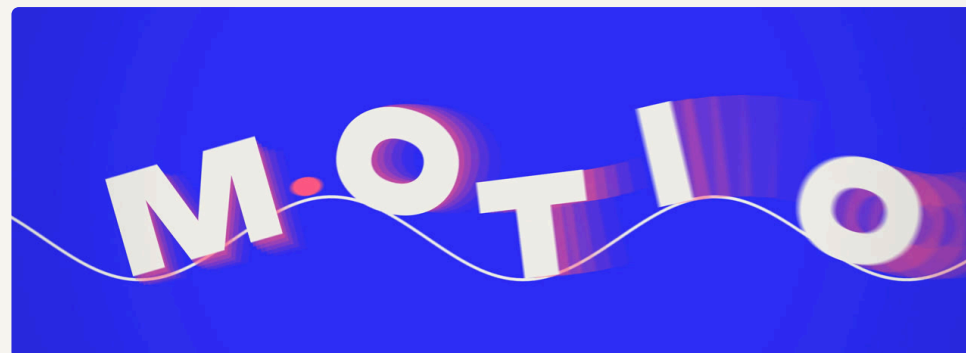
7.30 s the dot wedged in the A's counter • 9.85 s ultramarine iris opening from the dot

Picture. A blueprint grid draws out from the centre, labelled like a type specimen (CAP 360, MID 540, BASE 720, STROKE 76). Then M, A, K and E are drawn one per beat as continuous pen strokes. The dot drops into the A where the crossbar should be. Its resting height was solved so it wedges exactly between the A's inner edges. At 8.5 s it crouches and hops over the K, clearing it by a hair (about ten pixels), onto the tip of E's middle bar. That bar then shoots out to become the horizon while everything else retracts, and an ultramarine iris opens from the dot.

Sound. Ticking hats as the grid draws, woodblock clacks tuned upward per glyph, a two-part clink for the dot, a sine hop and boop, a zip for the line and a riser into the cut.

STAGING RHYTHM APPEAL MATCH CUT

● 0:10-0:16 • MOTION



11.00 s letters riding the wave with echoes • 15.40 s the build: tilt, zoom, shear, speed

Picture. The horizon comes alive as a travelling sine wave; the dot surfs it leftward and hops out. The letters of MOTION follow, sitting on the wave and rotating to its tangent. They land on consecutive sixteenths, then the wave flattens and they settle one after another. At 12.0 s: the slam. Six marquee rows open like blinds and scroll in alternating directions, with speed spiking on every kick. From 14.0 s the speed doubles every half-second while the camera tilts and pushes in and the rows shear with their own velocity. At 15.75 s everything decelerates like tape and collapses to a single line.

Sound. Impact and crash, full groove, a C-D-E-G-A-C arpeggio as the letters land, the slam, a snare roll from 8ths to 16ths to 32nds under a riser, and a tape-stop pitch dive into silence.

PATH ANIMATION OVERLAPPING ACTION REPETITION EXAGGERATION

● 0:16-0:18 • WITHOUT



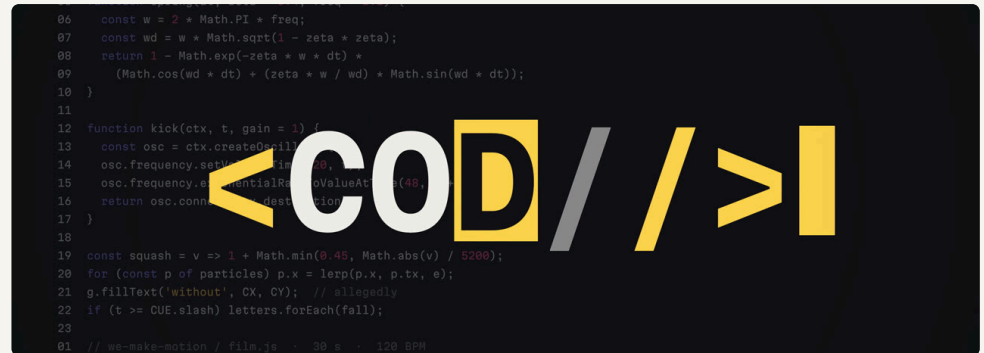
17.20 s the dot is the tittle of the i • 17.80 s diving into the ink of the o

Picture. A hard cut to chalk and stillness. The line from MOTION switches off like an old television. “without” resolves letter by letter from blur, with its tracking tightening. The *i*’s own tittle is masked out and the dot drops in to take its place. To find the tittle, the renderer rasterises the glyph and flood-fills its top-most blob. Then the camera dives into the *o*. The zoom target is the deepest point inside the glyph’s ink, found with a distance transform, so the stroke fills the screen and becomes the next scene’s ground.

Sound. The drums drop out completely. There is one glass ping (E6 with inharmonic partials), some air, three fading clacks as the dot lands and bounces, and a riser into the cut. Nearly two seconds of quiet make the next hit land harder by contrast.

CONTRAST SILENCE AS DESIGN TRANSITION THROUGH FORM

● 0:18-0:22 • CODE



19.60 s decoding, selection highlight • 20.60 s struck through and falling

Picture. The backdrop is a syntax-highlighted editor showing snippets modelled on the film’s own source, plus one invented joke line: `g.fillText('without', CX, CY); // allegedly`. The brackets spring in, the letters scramble and lock on consecutive eighths, each flashing a butter selection highlight, and a caret blinks on the beat. At 20.0 s a flamingo bar slashes through. Every glyph falls with its own spin, and the lines of code drop away behind them, each at its own moment. The bar hangs in mid-air for a beat, in the cartoon tradition, before straightening and exploding into a two-layer wipe.

Sound. Boom and glitch burst, bit-crushed hats, square-wave chatter that thins as letters lock, four rising locks, a slash, seven falling bleeps and two whooshes for the wipe.

RHYTHM IMPACT GRAVITY COMIC TIMING LAYERED WIPE

• 0:22–0:26 • THE CLAIM



23.10 s MOTION slams in, stretched and sheared by its own speed •
25.80 s the lockup

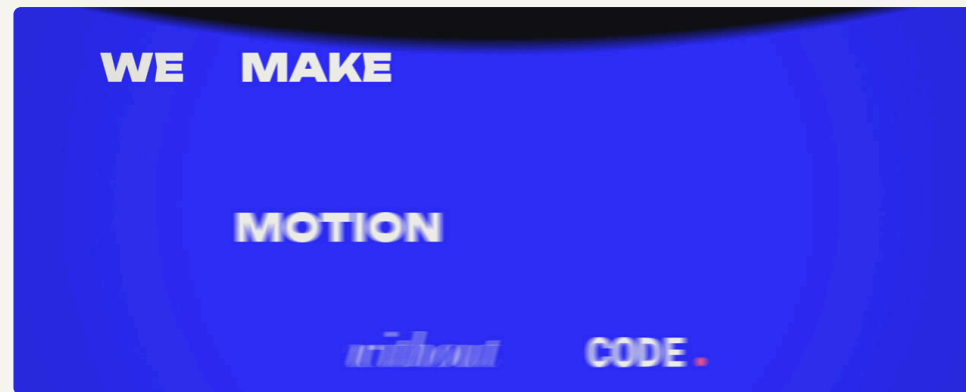
Picture. A Swiss-style stacked lockup, and each word gets its own verb on its own beat. WE and MAKE rise through masks, letter by letter. MOTION slides in from off-screen, stretched and sheared in proportion to its speed. *without* whispers in from blur. CODE decodes. Then the dot drops from the top of frame and bounces into place as the full stop, with the same physics and timings as the opening.

Sound. Five hits: two snares with clacks, a crash and impact for MOTION, the glass ping again for *without*, a glitch for CODE. Then the opening drum fill plays again, note for note, as the dot bounces.

HIERARCHY

CALLBACK

• 0:26–0:30 • THE FINE PRINT



26.40 s FLIP reflow under a falling ink curtain • 28.80 s the
asterisk and its footnote

Picture. Every word travels from its lockup position and size to its place in a single-line sentence, on alternating arcs with a small back-ease pull, while a liquid ink curtain falls. At 27.25 s the full stop rises to superscript and grows six arms: it is now an asterisk. The footnote types itself at 48 characters a second: “**Except this one. This one is 1,941 lines of code.*” Then everything fades, and the asterisk drifts to centre, retracts its arms, turns back into the opening dot and blinks out.

Sound. A reflow whoosh, a sproing (a frequency-modulated sine) for the asterisk, one tick per typed character (spaces are silent), a soft final kick with a low glass tone, and the opening blip to close.

FLIP REFLOW

MORPH

BOOKEND

A drum kit made of maths

VOICE	RECIPE	USED FOR
Kick	Sine swept 170 → 46 Hz in 90 ms, exponential decay, 2.5 kHz noise click	Beats, bounces (pitch rises per bounce)
Snare	Band-passed noise at 1.9 kHz + triangle 200 → 150 Hz body	Backbeat, the build's roll
Clap	Three noise bursts 11 ms apart through a 1.2 kHz band-pass, then a tail	The WE lock, slam, slash, backbeats
Hats	High-passed noise, 35 ms closed / 240 ms open; bit-crushed in CODE	16th-note motor
Clack	Sine + triangle at 2.71× (woodblock), low sine thump underneath	MAKE glyphs, tittle, bounces
Pluck	Triangle through a closing low-pass into a ping-pong delay, plus a quiet octave sine	Mitosis, the MOTION arpeggio
Glass	Partials at 1, 2.76, 5.4, 8.93× with staggered decays, heavy reverb	without, the claim, the ending
Whoosh	Noise through a swept band-pass with a stereo pan ramp	Exits, wipes, slides
Riser	Band-pass sweep 300 Hz → 9 kHz + low-passed saw rising three octaves	Into WE, MOTION, the slam, the collapse, CODE
Impact	Sub sweep 70 → 26 Hz, low-passed noise, kick	Scene arrivals
Tape stop	Saw chord dropping to 8% pitch under a closing filter	The collapse at 15.75 s
Glitch	Random square blips every ~17–42 ms through an 11-level bit-crusher	CODE arrivals and slash
Sproing	Sine with a 22 Hz vibrato whose depth decays from 120 Hz to 2 Hz	The asterisk
Tick	12 ms band-passed noise click + a tiny sine	Grid lines, typewriter

SIGNAL CHAIN

686 voices → pan → dry / reverb send / delay send

Convolution reverb 2.8 s generated impulse, deterministic noise ^3.4 decay

Ping-pong delay dotted eighth (375 ms), 38% feedback, 3.2 kHz low-pass

Drive +2.9 dB → bus compressor -16 dB, 3.5:1, 4 ms / 200 ms

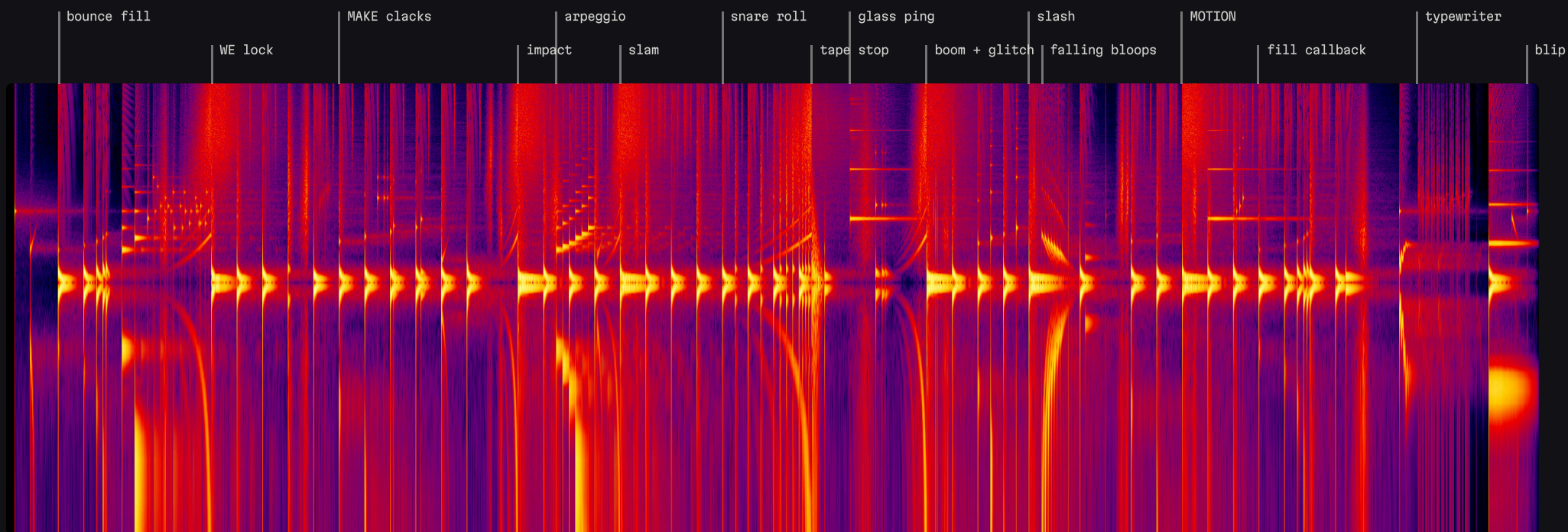
Limiter -2 dB, 20:1, 1 ms → master fade 29.55–30 s

The score renders in an `OfflineAudioContext` faster than real time. The browser player plays that buffer and slaves its picture clock to the audio clock, compensating for output latency. The offline renderer runs the same function in headless Chromium and exports a 48 kHz, 16-bit WAV, which the MP4 carries as 256 kb/s AAC. Same score, same code; renders differ only at about -133 dB, the noise floor of the audio engine.

Plucks, blips and glass pings use C major pentatonic notes, so accents from different scenes sit together. The kicks' tails are the only bass; they, and the clacks, are tuned by ear rather than to the scale.

Reading the score back

I can't listen to audio, so I checked it the way an engineer would: loudness measurement and a spectrogram of the rendered WAV read against the cue sheet. Time runs left to right across 30 s; frequency is ffmpeg's log scale up to 16 kHz, most legible in its upper half.



Visible as designed: the accelerating bounce fill at 1–2 s; the arpeggio staircase around 10.8–11.4 s; the snare roll thickening into the tape-stop dive at 15.75 s.

The near-empty band at 16–18 s is the silence, with the glass ping ringing through it as a horizontal line. The falling bleeps at 20.25 s show as descending diagonals.

Loudness: first pass -17.0 LUFS, a little shy, so I added bus drive (-15.9). Charting the energy curve for this document then showed the “silence” peaking at -9.8 LUFS, because the glass ping sits where hearing is most sensitive. The pings came down 5–11 dB. Final: -16.4 LUFS integrated, -0.8 dBFS peak, 3.3 LU range.

From source to 4K master

src/*.js
core, scenes, audio, player
→ one index.html (1,941 lines)

Headless Chromium ×3
the same page, fonts inlined, canvas at 3840×2160

24 sub-frames / frame
summed in 32-bit float, one PNG per frame

1-second segments
each piped straight into its own ffmpeg; no frame files

Lossless concat
+ AAC soundtrack; 1080p is a Lanczos downscale of the 4K

REAL MOTION BLUR

A 180° shutter at 60 fps is open for 1/120 s. For every frame the renderer draws the scene at 24 evenly spaced instants inside that window and averages them in 32-bit float. Fast elements smear as a camera would record them, slow ones stay sharp, and static ones stay unblurred. That is 43,200 full scene renders for 1,800 frames. In the browser, where there is no budget for that, the particle scene fakes blur with velocity streaks instead (`R0.streaks`).

RENDERED, NOT UPSCALED

Everything is drawn in a 1920×1080 design space and scaled to the canvas, so the 4K master is the same code on a canvas twice the size. A 4K frame is about 15 MB of PNG, so frames never touch disk. Workers pull one-second segments from a queue and pipe them into per-segment encoders, and the segments are stitched without re-encoding. Any second can be re-rendered on its own, which is how QA fixes were patched in. At 4K one new flaw surfaced: hairline seams inside letters where the variable font’s contours overlap. Large glyphs are now also stroked in their own colour at about one device pixel, which closes them.

NUMBERS

Frames · sub-frames	1,800 · 43,200
Shutter	180° (1/120 s), 24 samples
Workers	3 headless Chromium pages, 4 cores
4K render wall-clock	72 min
4K video	H.264 High 5.2, CRF 17, light grain
1080p video	Lanczos downscale, CRF 18
Audio	48 kHz WAV → AAC 256 kb/s
Masters	4K 39.5 MB · 1080p 15.9 MB

DETERMINISM IS WHAT MAKES IT WORK

No frame depends on the one before it, so any worker can render any second in any order, and a single second can be replaced later. Fonts are embedded as data so every worker lays out identical glyphs.

How it got made, in order

STAGE	WHAT HAPPENED
00 Brief	Read the brief, wrote the self-prompt and the rules on page 3, and planned all eight scenes on the beat grid on paper before writing any code: every hit, entrance and cut given a timestamp.
01 Toolchain	Found headless Chromium and Playwright in the environment. Installed an ffmpeg build with x264 and AAC. Downloaded the three typefaces so offline renders lay out identically to the browser.
02 Engine	Wrote the core: frame and tempo constants, the palette, easing library, analytic spring, keyframed ballistics, hash noise, the cue sheet, the camera-impact list, and glyph analysis (rasterise, distance transform, flood fill).
03 Scenes	Built the eight scenes in three passes (drop, WE and MAKE; MOTION and without; CODE, the claim and the fine print), then the HUD, glitch pass, compositor and motion-blur accumulator.
04 Score	Synthesised the kit (18 voice types) and wrote the score against the same cue sheet, then set up the mix bus: reverb, delay, bit-crusher, compressor and limiter.
05 Player & renderer	Wrote the in-browser player (audio-clocked playback, scrubber, dope sheet, principles overlay), a build step that stitches the sources into one page, and the offline renderer.

STAGE	WHAT HAPPENED
06 First look	Rendered 48 stills across the timeline as two contact sheets (next page). The whole film worked end to end on the first render. Review found two picture bugs and one choreography change.
07 Full-size pass	Inspected frames at 1:1. Found the variable font’s overlapping contours showing in the outline rows, and the HUD colliding with the marquee. Fixed both.
08 Sound check	Measured loudness (−17.0 LUFS, a bit shy) and added bus drive (−15.9 LUFS). Verified every cue against a spectrogram.
09 Blur test	An 8-sample blur stepped visibly on the fastest move (MOTION entering at about 17,000 px/s). Rebuilt the accumulator in float at 24 samples.
10 UI pass	Screenshotted the player at desktop and phone widths. Fixed a hidden-element rule, ambiguous timecodes, colliding scrubber labels and an oversized play button on phones. The player now idles instead of redrawing when paused.
11 Master	Published the player. Rendered 1,800 frames with 24-sample blur and muxed the master. Re-shot the last 25 s so the footnote quotes the final line count. The first encode was 353 MB because film grain is incompressible noise. Lighter grain at CRF 18 brought it to about 16 MB with no visible loss.
12 Mix fix	Charting loudness for this document showed the “silence” at 16 s peaking at −9.8 LUFS because of the glass ping. Lowered the pings 5–11 dB. A renderer bug wiped the frame cache during the audio-only pass, so the master was re-rendered in full. The frames are deterministic, so they came back identical.
13 This document	Laid out in HTML in the film’s own typefaces, printed to PDF with the same headless browser, then proofed page by page.



First-pass contact sheet, 0.3–17.4 s, one frame per tile, rendered before any fixes. The rows at 12.4 and 13.2 s show the pink-fill bug.

First render, first notes

The first full render already worked end to end: every scene, transition and cue in place. The review notes were about craft and one real bug.

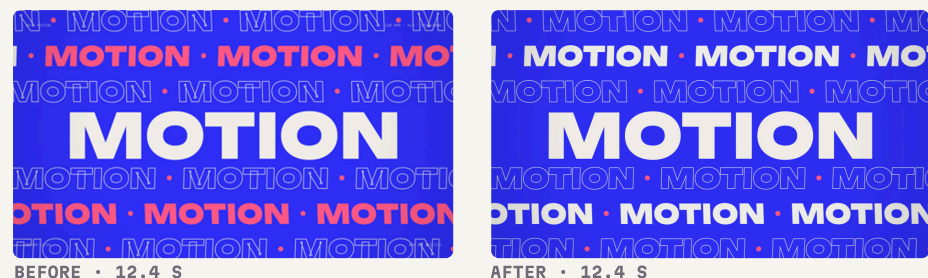
BUG Filled marquee rows came out flamingo. The helper that draws the separator dots left its fill colour set, so every word after the first dot inherited it. Fixed by setting the fill per word. Pink rows also broke Rule 7: flamingo is for the dot and its accents, not whole rows of type.

TIMING The collapse used an exponential ease-in, so the rows didn't visibly move until the last three frames. Switched to a quartic ease-in so the viewer sees the squeeze happen.

CHOREOGRAPHY On the back-half sheet, the ending blinked out at the right-hand end of the sentence. The film opens on a centred dot, so it should close on one. The asterisk now drifts home, turns back into the opening dot, and blinks out.

KEPT The particle swarm, the MAKE construction and the zoom into the ink worked as planned and were left alone.

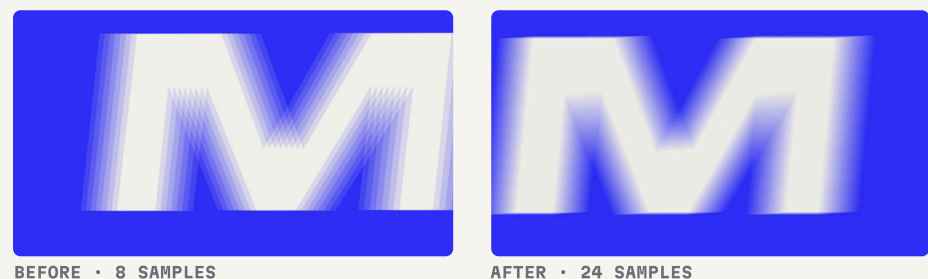
Before and after



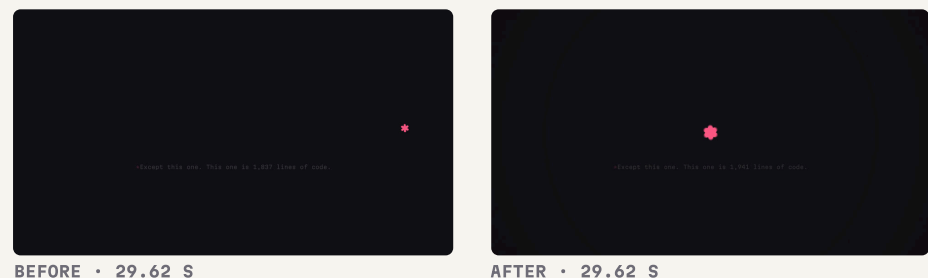
Leaking fill colour. The helper that draws each flamingo separator dot changed the fill colour and never changed it back, so every word after the first dot turned pink. Now each word sets its own fill. The before frame is a thumbnail from the first contact sheet.



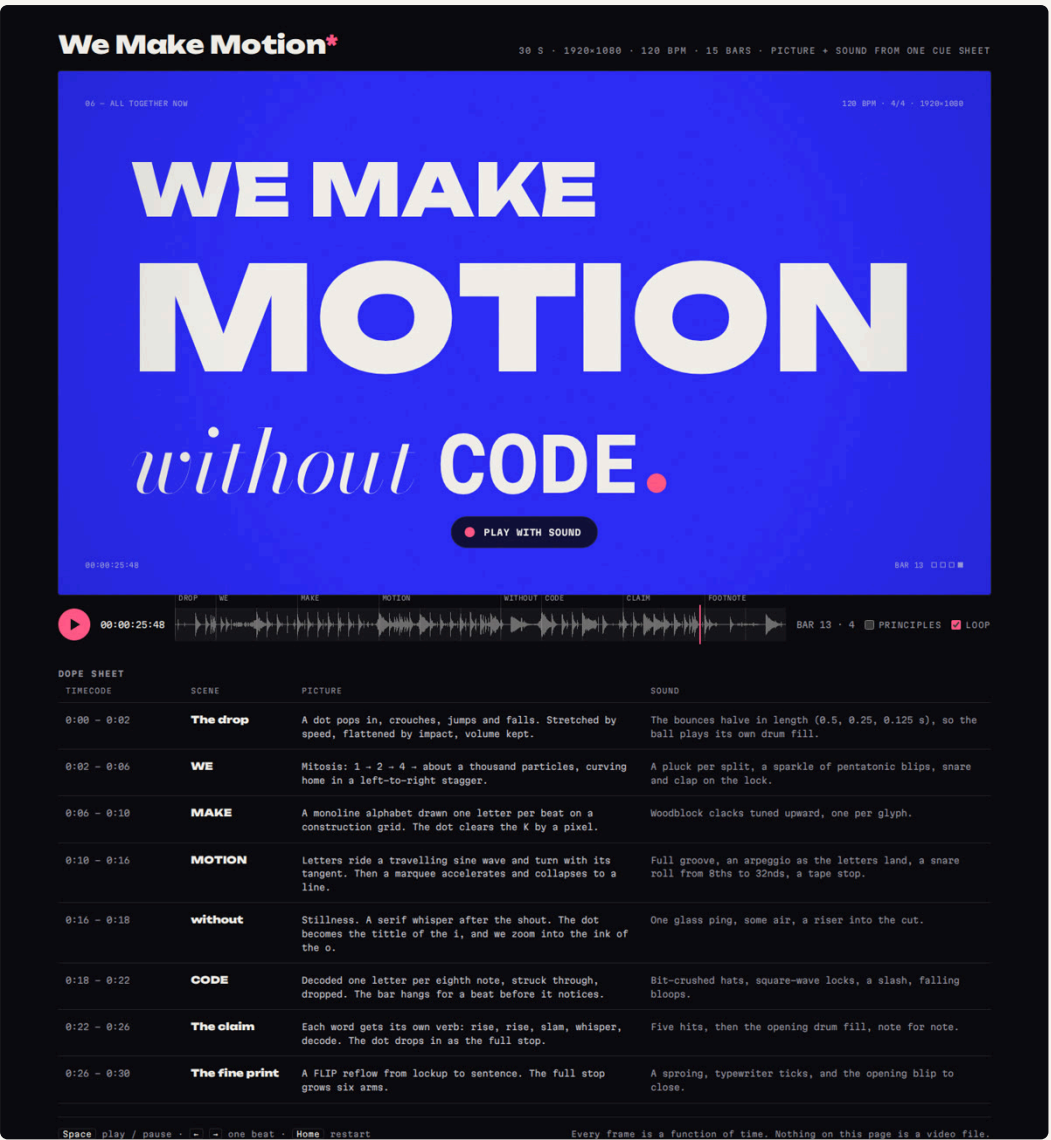
Overlapping contours. Variable fonts build letters from overlapping shapes. Filled, that's invisible; stroked, the overlaps show as stray lines inside M, N and T. The fix strokes at double width, then fills with the ground colour so only the outer half of each stroke survives.



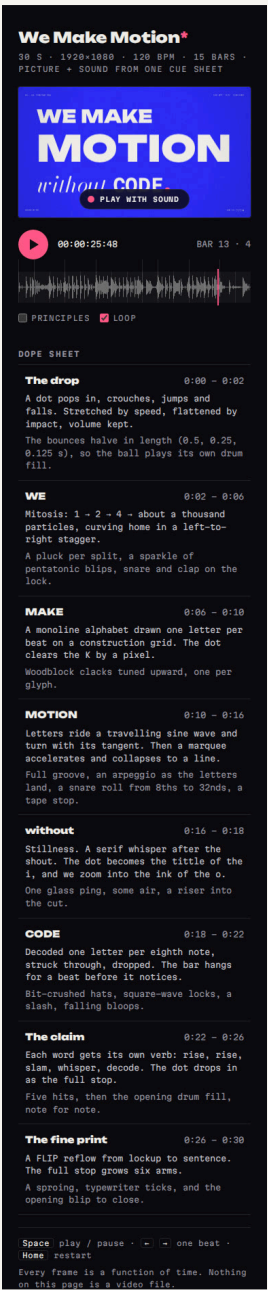
Stepped motion blur. MOTION enters at roughly 17,000 px/s, about 140 px of travel inside one shutter. Eight samples left visible copies. Twenty-four samples, summed in 32-bit float so the average doesn't band, give a continuous smear.



The bookend. The asterisk used to blink out where the sentence ended, off to the right. The film opens on a centred dot, so now the asterisk drifts home, shrinks back to the opening dot's size, and blinks out with the same blip it started with.



The published player at 1440 px, paused on the poster frame.



At 390 px.

The player

The browser version is not a video. It renders the film live on a canvas sized to the player and the screen's pixel density (up to 2^x, at most 1920 px wide), with the soundtrack synthesised on load. Picture is slaved to the audio clock, so it stays in sync.

TRANSPORT Play, SMPTE timecode, bar and beat, a scrubber drawn from the real waveform with bar lines and scene marks.

DOPE SHEET Every scene with its picture and sound notes; click to jump.

PRINCIPLES An overlay that names the technique on screen as it happens (29 captions).

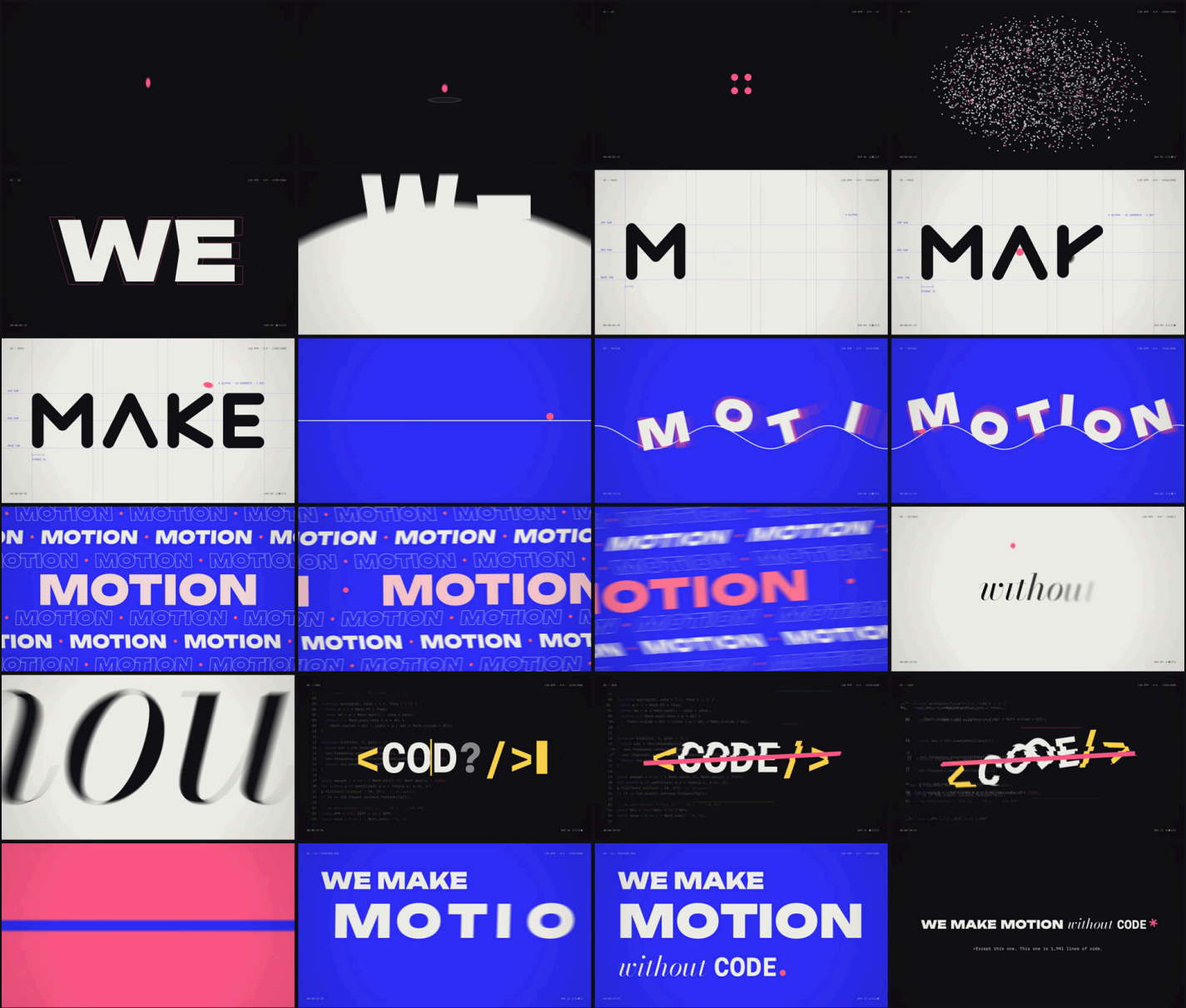
KEYS Space, ← → by one beat, Home.

UI FIXES FROM REVIEW

Hidden elements forced hidden; dope-sheet timecodes changed from SS:FF to m:ss; short scrubber labels; a smaller play button on phones; the canvas idles when paused.

The final cut, in 24 frames

FROM THE MASTER • 24-SAMPLE MOTION BLUR



Same film, new medium

The follow-up brief: a 3D version with glowing accents, grit textures, and interesting mattes and masking.

I kept the cut, the cue sheet and the score, and swapped the renderer. The 3D edition calls the 2D film’s own choreography functions (the dot’s ballistics, the particle paths, the letters on the wave, the falling glyphs) and re-stages them in WebGL. Every hit still lands where it did, because it is the same maths.

What changes is material. Letters are extruded from the film’s fonts and lit. Accents are emissive and bloom. Surfaces carry procedural grit. And every transition but one is now a matte (MOTION still hard-cuts into *without*): a shape that decides where one world ends and the next begins, with a noise-bitten, glowing seam.



13.20 s: extruded MOTION in front of marquee rows placed at different depths.

MOMENT	3D TECHNIQUE
The drop	The dot is an emissive sphere carrying its own light. The gritty concrete floor only exists where it lands.
WE	1,897 glowing particles scatter in depth, flatten onto the letters’ face, and the word extrudes its thickness on a spring while the camera orbits.
→ MAKE	Liquid wipe as a noise-edged matte with a glowing seam.
MAKE	Glossy ink tube strokes (cylinders with spherical joints) drawn on like a pen, casting shadows on a paper wall.
→ MOTION	Iris matte opening out of the dot.
MOTION	Letters ride a glowing ribbon with echo trails in depth; a dolly zoom stretches the marquee world during the build.
without	Bodoni surfacing out of fog. The o’s counter is a portal matte into CODE, and the camera flies through it.
CODE	Three parallax layers of source; butter-glow extruded brackets; letters tumbling toward the lens.
The claim	WE and MAKE rise through clipping-plane mattes at their baselines; MOTION slides, <i>without</i> surfaces, CODE decodes; a spotlight sweeps the lockup.
Fine print	One world rendered in two looks, joined by a noise-edged curtain; the full stop grows six glowing arms.

Glow, grit and mattes, in order

Scene A + B

HDR, 4× MSAA, shadows, fog

Matte

3D mesh or 2D-drawn shape, softened

Composite

fbm noise bites the edge; seam glows

Bloom

threshold 1.35, 6-level dual Kawase

Grade

hue-safe tone curve, CA, vignette, glitch, HUD

Finish

sub-frame average, grain, dust, scratches, weave

MATTES WITH TEETH

A matte decides, per pixel, whether you see world A or world B. Mine come from two places: real 3D geometry rendered in white (the counter of the o, seen through the moving camera) or shapes drawn on a flat canvas with the 2D film's helpers (the liquid wipe, the iris, the band). The mask is softened, then pushed around by fractal noise before it is thresholded. That gives a torn, gritty edge instead of a clean line. The same noise drives a glow along the seam, so every transition burns rather than cuts.

```
k = mask + (fbm(uv) - .5) * edgeNoise
mm = smoothstep(.5 - w, .5 + w, k)
col = mix(A, B, mm) + glow * exp(-((k - .5)/2.2w)2)
```

GLOW WITHOUT FOG

Only emissive things should glow: the dot, the particles, the ribbon, the brackets, the blade. The bloom threshold sits at 1.35 in linear light, above anything a lit surface reaches, so chalk letters stay crisp while accents flare. Bloom is a six-level dual-Kawase pyramid: cheap, smooth and wide.

COLOUR YOU CAN TRUST

The standard filmic curve (ACES) pushes saturated blue toward purple, which is wrong for a brand ultramarine. The grade uses a luminance-only curve instead: linear to 0.8, a soft shoulder above, and a gentle roll to white only for hot emissive cores. #2D2DF5 stays #2D2DF5.

GRIT, THREE WAYS

Surfaces: procedural concrete and paper textures (value noise, specks, scratches) drive roughness and bump, so raking light finds texture. **Film:** luminance-weighted grain, sparse dust specks, a rare scratch and a sub-pixel gate weave, all applied after motion blur so they don't average away. **Signal:** chromatic aberration that widens during the build, and glitch slices on the CODE hits.

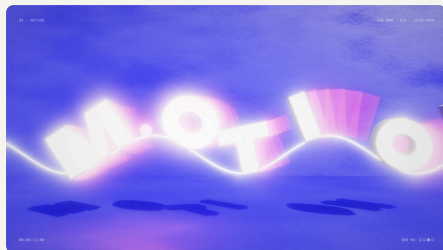
TYPE AS GEOMETRY

Glyph outlines come straight from the font files via opentype.js, are unioned with a polygon clipper, then extruded with a bevel. Faces and sides take different materials.

BUILT FOR 4K

The first 4K test showed that film effects written in device pixels don't scale: grain striped where its hash ran out of float precision, dust quadrupled, and bloom and matte edges tightened. The finish now works in 1920×1080 design pixels, bloom and matte softness keep their reach relative to the frame, and text textures are sized to the output. Downscaled to 1080p, a 4K frame differs from a native 1080p render by under 2 grey levels on average.

3D, before and after

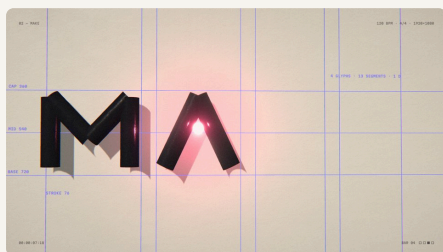


FIRST PASS • 11.0 S



FINAL • 11.0 S

Overcooked. The first 3D render worked end to end, and looked like a nightclub in fog. The bloom threshold (1.0) caught every lit surface, and ACES turned ultramarine lavender. The fix was a 1.35 threshold, a hue-preserving tone curve and lights at roughly half power. The same pass thinned the dust, which read as stars.



SECOND PASS • 4.4 S

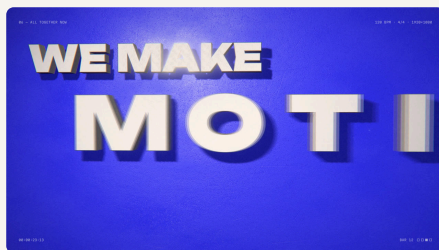


FINAL • 4.4 S

Seams in the letters. Unbounded's static instance still stores letters as overlapping pieces. Bevelled one by one, the pieces show their edges across the face of W and E. Unioning each glyph's contours before extruding gives one clean outline and one clean bevel.



THIRD PASS • 23.2 S



FINAL • 23.2 S

The lockup re-lit. A hard spotlight from far left blew a hotspot into the wall, and its long shadows read as a second, dark copy of the words. Moving the key frontal and wide, and the wall closer, gives tight contact shadows and an even, gritty ultramarine.



FOURTH PASS • 16.9 S

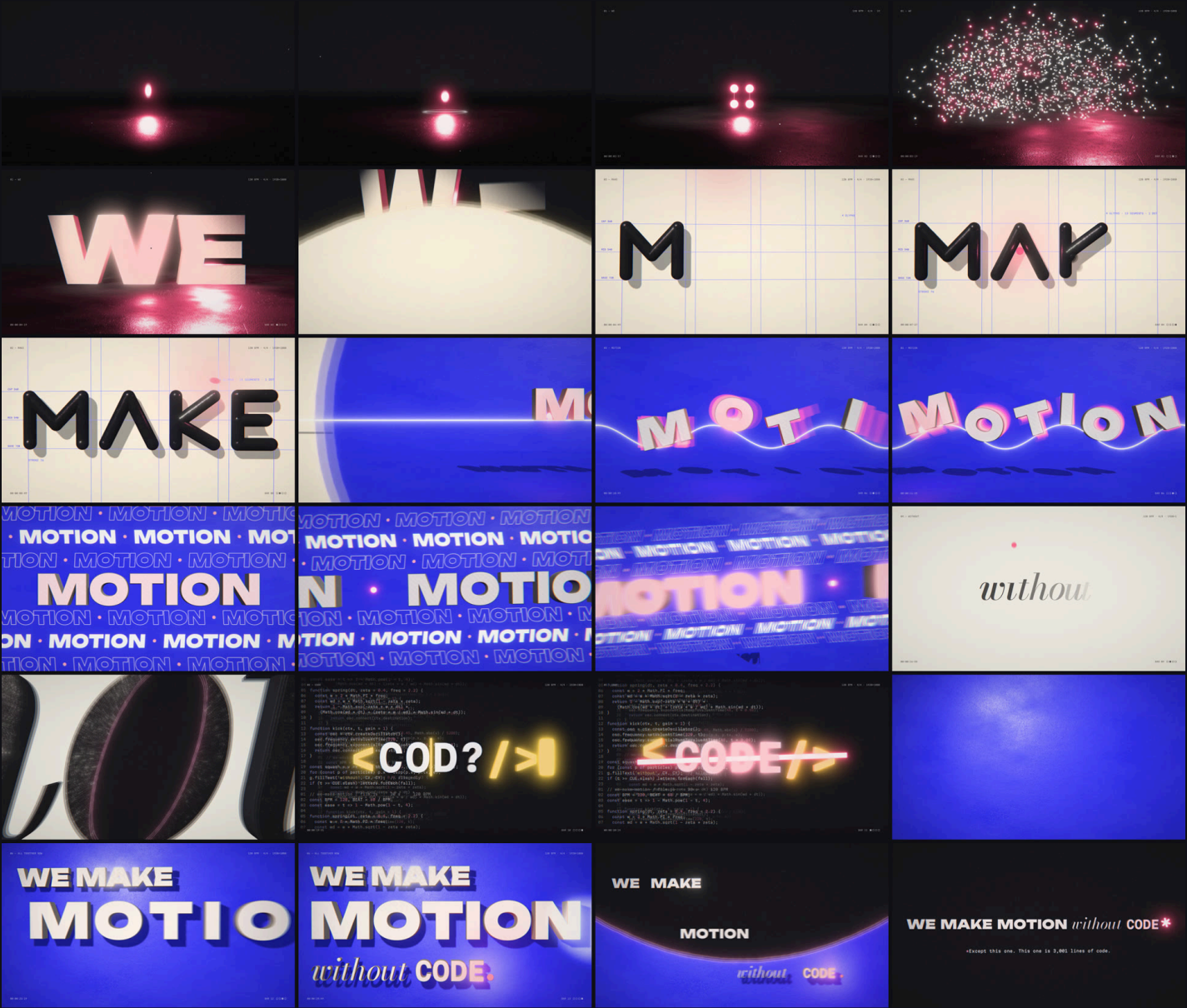


FINAL • 16.9 S

A dot, not a ring. On light grounds, a hot emissive core rolls to white and vanishes into the paper, leaving a pink ring. Light scenes now use a solid flamingo at 1.05x, below the bloom threshold, for the dot; dark scenes keep the glow.

The 3D cut, in 24 frames

1,800 FRAMES • 5,400 SUB-FRAMES • 334 MIN AT 4K ON SOFTWARE WEBGL



Review at scale

Each edition is 1,800 frames. I can't sit back and watch a film play, so review became a pipeline: independent reviewers each take a slice, and every finding has to survive a skeptic before anyone fixes it.

300

FRAMES INSPECTED

150 per edition, one every 0.2 s, split across 12 inspectors in five-second windows.

22 / 46

FINDINGS CONFIRMED

One skeptic per finding, told to refute it unless it was visible, unintended and noticeable at speed.

3

ROUNDS OF FIXES

22 verifiers compared old and new frames at the same timecodes; 2 sweeps hunted for regressions.

43 / 520

CLAIMS CORRECTED

Every technical claim in this document, checked against the source code.

HOW A FINDING SURVIVES

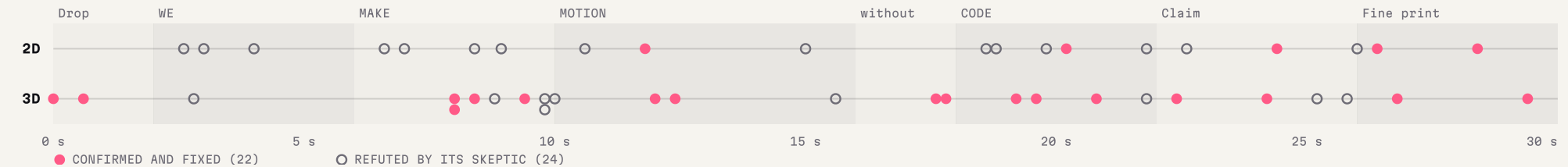
Inspectors got the design intent up front, so deliberate choices (a hard cut, a glitch on a hit) weren't reported, and every finding had to cite its frames. A skeptic then opened those frames plus the neighbours 0.2 s either side. It confirmed only if the flaw was visible, unintended and noticeable at full speed; if in doubt, it refuted. More than half the findings, 24 of 46, died there. Most were transitions that look odd for a frame or two, or effects working as designed.

FIXES GET CHECKED TOO

Each fix was re-rendered at the same timecodes and handed to a fresh verifier with before and after. Round one: 17 fixed, 4 partly fixed, 1 not fixed, and the sweeps flagged 13 more. The misses taught the most. Loosening MOTION's tracking only shrank a letter collision from 0.65 s to 0.2 s. What worked was a real collision pass (next page), proved by drawing each letter into its own mask: 103 frames, zero overlapping pixels. I checked rounds two and three myself, on fresh renders.

THE DOCUMENT, AUDITED

520 claims were checked against the code; 52 were flagged, and 43 still stood after a second agent tried to defend the original wording. Where the text was right in spirit but the code disagreed, I changed the code: wipes, camera impacts and glitches now derive from the cue sheet, as the timing page says. That refactor had to change nothing on screen. 19 test frames came back pixel-identical, and the soundtrack differed by -132.8 dB RMS, level with the -132.7 dB between two identical renders.

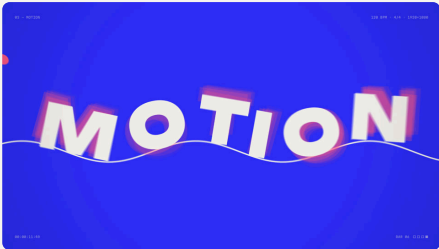


Every finding from the frame review, placed at its timecode. Most confirmed flaws were in the 3D edition, the newer and more complex of the two.

Caught, fixed, checked



BEFORE • 2D • 11.8 S



FINAL • 2D • 11.8 S

Letters that fused. Riding the wave, neighbours that leaned toward each other swept into one another. MO became one shape and IO read as IO. Now the row is laid out as a whole. Each letter's box is rotated with its lean, and the next letter is pushed until it clears it, then re-seated on the wave. The push fades with the wave, so the slam lands on the true spacing. In 3D the gap also widens by the width the extruded side walls project to.



BEFORE • 2D • 24.2 S



FINAL • 2D • 24.2 S

Grey on grey. CODE's selection highlights faded out with transparency, which mixed butter and ink into murky greys for several frames. They now wipe off at full strength, with an ink copy of each glyph clipped inside the moving box, like a text selection being released.

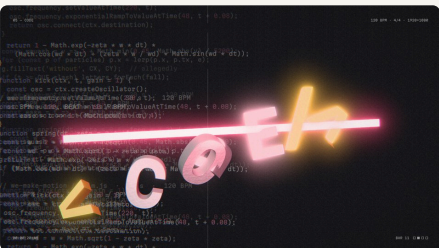


BEFORE • 3D • 8.0 S



FINAL • 3D • 8.0 S

Labels under the M. The MAKE guide labels lived in the 3D scene, so the M covered them as the camera drifted, and the ink tubes had lost their round joints. The labels are now drawn flat over the frame, each placed where its guide line lands on screen, and the joints are back.



BEFORE • 3D • 20.8 S



FINAL • 3D • 20.8 S

Crossed paths. The falling glyphs pivoted at their baselines and their paths crossed, so C, O and D merged mid-air. Each now pivots at its cap-height centre, fans out from the middle and falls at its own depth.

Appendix

FILES

<code>we-make-motion-4k.mp4</code>	2D master, 3840×2160
<code>we-make-motion.mp4</code>	2D, 1080p from the 4K
<code>../we-make-motion-3d/*.mp4</code>	3D, 4K and 1080p
<code>index.html</code>	The film and player in one page
<code>src/core.js</code>	Constants, maths, cue sheet, glyph tools
<code>src/scenes-1..3.js</code>	The eight scenes, HUD, compositor, blur
<code>src/audio.js</code>	Kit, score, mix, WAV encoder
<code>src/player.js</code>	Transport, dope sheet, captions
<code>build.mjs</code>	Stitches src into index.html
<code>render/render.cjs</code>	Offline renderer and mux
<code>docs/process/</code>	This document's source
<code>../we-make-motion-3d/</code>	The 3D edition: gl3, world3, scenes3, player3, render3
<code>../render4k.cjs</code>	Segment-streamed 4K renderer, both editions

REBUILD

```
node build.mjs
pip install imageio-ffmpeg
node ../render4k.cjs --edition 2d
# preview stills
node render/render.cjs --stills 4,12.5
# re-shoot a range, re-mux
node render/render.cjs --range 1650,1800
# the 3D edition
node ../render4k.cjs --edition 3d
# this document
node docs/process/build-pdf.cjs
```

CREDITS

Typefaces: Unbounded, Bodoni Moda and Martian Mono, all from Google Fonts under the SIL Open Font License. Everything else, including every sound, was generated.

LIMITS, HONESTLY

LISTENING I can't hear the mix. It was verified by measurement and spectrogram, so the balance deserves a pair of human ears.

LIVE PLAYBACK The browser version renders in real time, so very old phones may drop frames. The MP4 is the reference.

VARIABLE FONTS Canvas can't animate font axes smoothly, so weight and width changes were done with transforms instead.

3D The 3D master uses 3 motion-blur samples, not 24, because this machine renders WebGL in software. The live 3D player needs a reasonable GPU.

NEXT

8K masters from the same code (resolution isn't baked in), vertical and square cuts (which need re-layout rather than cropping), and a GPU render of the 3D edition at 24 samples.